Complete cold costs and matched compression

Status: independently replayed finite software-count diagnostic; benchmark admission remains incomplete.

The fixed plan contains 4749 profiles and 4749 attempted executions. There are 60 distinct public DLP targets in this cohort. Configurations and process repetitions are correlated; a recovered execution is not another independent target.

Recorded cohort and outcome	Executions
AA / recovered	6
archived-exhaustive-discovery / recovered	3
development-alpha / ambiguous	12
development-alpha / recovered	360
development-alpha / underfilled	108
development-cold / ambiguous	408
development-cold / recovered	756
development-cold / underfilled	36
development-reference / capped	3
development-reference / recovered	897
regression-60 / ambiguous	1026
regression-60 / recovered	1134

The ratios below use only successful matched recoveries, with every unsuccessful attempt retained in the table above. Endpoints are observed ranges, not confidence intervals.

Numerator / denominator	Median [min, max]; pairs
Full / compressed, cold	0.94084 [0.65134, 0.99974], n=1125
Full / compressed, matrix	0.47265 [0.17979, 1.1047], n=1848
Ordinary / twist, cold	0.98560 [0.55827, 1.2956], n=750
Map / scalar, cold	0.99258 [0.98633, 0.99645], n=180
IC / strongest completed measured reference	59.650 [7.3906, 8071.0], n=2253

Ratios below 1 favor the numerator. Compression cost more in every successful matched cold pair in this implementation. The map/scalar saving is available to ordinary bases too. The ordinary/twist ratios do not establish a consistent twist-specific advantage. Five references share each Q: native rho, negation/cycle-escape rho, BSGS, negation BSGS and GP elllog. The strongest completed member is not a global fastest-rho certificate.

The60-target collector regression has 720 matched groups. Group, native full-root and GP full-root methods agree on the complete problem and outcome. Three reference executions hit their fixed algorithm cap; they are retained.

All raw instruction sums and completed source/map/full-fiber computations pass independent native/GP checks. Initial GP references were strictly rejected for undeclared automatic stack-growth warnings. Their unchanged-script successors use a larger initial stack and pass the same strict policy. No attack profile was repeated.

The original summary stopped on identical missing query hashes in pre-query failures. A versioned post-data guard correction preserves successful-pair eligibility and formulas. Both versions remain. The pre-data machine protocol also encoded the explicit flat plan in fields the canonical checker treats as a rectangular product. Export schema checks pass, but full protocol/export admission fails (including the analysis-source correction). These are descriptive diagnostics, not promoted benchmark or leaderboard rows. The exact pre-data plan is complete; the checker finding is not evidence of missing planned profiles.

The unit is single-event guest instructions Ir, including actual resident-GP instructions in each exclusive phase. Cold setup, queries, complete decomposition, signs/deduplication, matrix work and source-point verification are charged per target. Input/program loading and result/trace serialization are excluded. No clock, cycle or primitive/GAE conversion is used.

Conditional affine Lean lemmas retain their stated hypotheses. Concrete arithmetic, rank/kernel spans and map fidelity are checked externally; a fully formal concrete bridge is open. Literature and access gaps are in the frozen protocol.

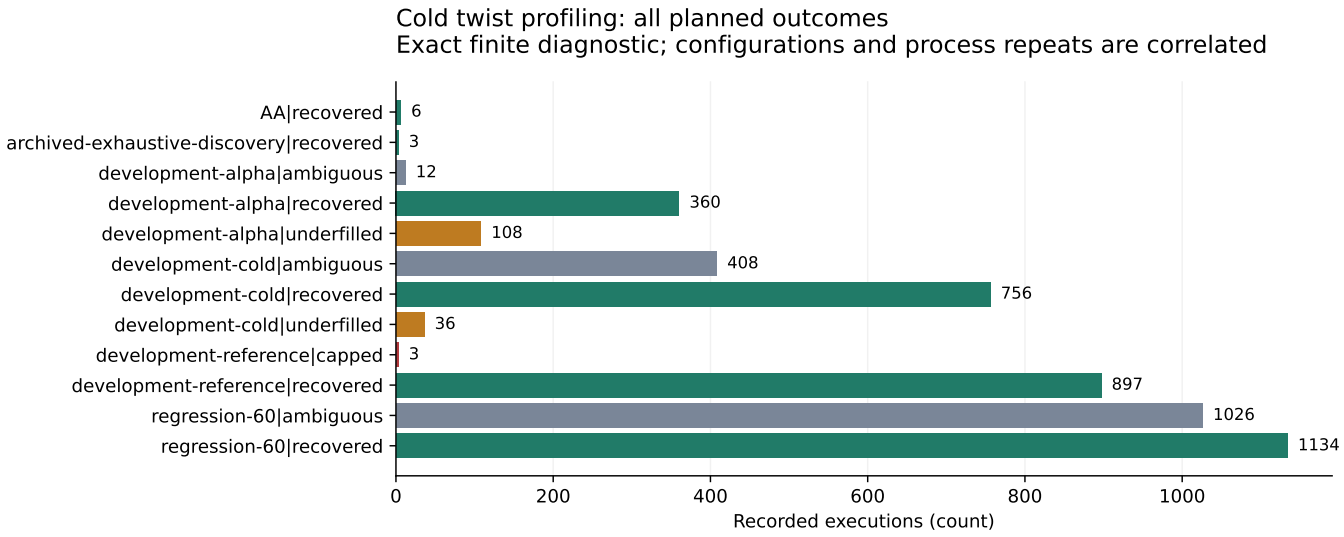


Figure 1: All recorded outcomes

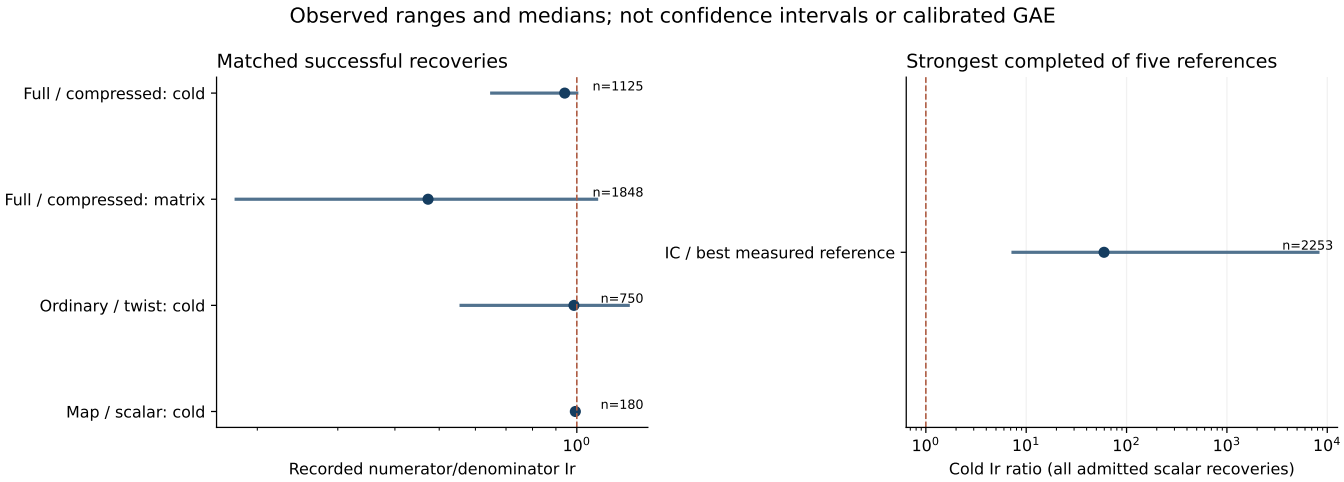


Figure 2: Native recorded ratio ranges

Sources: frozen protocol, native data manifest, source/byte pins. Independent GP admission, reporting correction.

Reproduction and raw-object manifests are linked from the evidence directory. The user-confirmed store is `setec-astronomy-artifacts`, region us-east-2. No NIST, asymptotic, novelty or calibrated-GAE result is established.